Dangers imposed to and by GenAI and possible countermeasures

Jannis Liebscher
HTWG Konstanz
Konstanz, Germany
jannis-liebscher@gmx.de

***Abstract*—This paper shows the different angels by which generative AI (GenAI) can be used for malicious causes and discusses some preventive measures**

I. Introduction

GenAI poses threats in many forms. On one hand, it can be used directly as a tool to serve a malicious purpose. On the other hand, GenAI itself can be the subject of an attack. Beyond that, an integrated AI can be exploited to enable attacks on systems connected with that AI. Both individually and in combination, these attack vectors represent threats that require mitigation.

II. Gen AI as a Tool

As is the case with all new technology, Generative AI is being exploited by malicious actors for malevolent purposes. In addition to this AI is a valuable target. Many companies start to integrate AI into their products and systems [8]. For that reason finding weak points and vulnerabilities which can be exploited represents an attractive target.

A. Phishing and Social Engineering

Large Language Models (LLMs) can be leveraged by adversaries to assist in the creation of spam and phishing emails [1]. Their use can significantly accelerate the generation of such content. Furthermore, these models are capable of producing highly convincing text that mimics human writing, substantially improving the quality of malicious messages. In scenarios where attackers face language barriers, LLMs can eliminate the linguistic inconsistencies that would normally raise suspicion. An attacker targeting users in multiple countries could generate emails in contextually appropriate French, Japanese, or Arabic without requiring a native speaker.

LLMs also enable automated personalization of phishing messages. By providing the model with publicly available information, such as social media posts, professional profiles, or past communications, an attacker can generate emails that reference the recipient's projects, colleagues, or interests. A typical workflow may include scraping data from LinkedIn, summarizing relevant activities, and generating an email that appears to originate from a trusted contact, thereby increasing the likelihood of successfully compromising the target.

B. Code Generation and Ransomware

Another concerning application is the use of LLMs for malware development [1]. Modern language models with coding capabilities enable even inexperienced users to generate functional ransomware or exploit scripts [6]. For example, an individual could request a simple script that encrypts a target directory and produces a ransom note. More advanced attackers can utilize these models to streamline the development process, automatically generating variations of malware to bypass signature-based detection systems. This accelerates the creation and deployment of new threats while lowering the technical barriers traditionally required for malware development.

C. Visual Misinformation and Manipulation

Visual Language Models (VLMs) further expand the range of malicious applications [1]. These models can generate highly realistic images or videos from textual prompts, enabling adversaries to fabricate convincing but entirely false visual content. Such capabilities can be exploited in misinformation campaigns, deepfake generation, and targeted harassment. For instance, a malicious actor could create an image of a political figure making a controversial statement or fabricate evidence of an event that never occurred.

The speed and realism achievable with VLMs allows for the rapid production of large volumes of such content, potentially overwhelming traditional moderation mechanisms. The combination of LLMs and VLMs could amplify this problem. In some scenarios, a fabricated news article may be paired with supporting images or videos, making detection and verification substantially more difficult for both human reviewers and automated systems.

III. Gen AI as a Target

Generative AI, when integrated into other systems, can become a target of attacks. In particular, AI agents that interact with such systems with higher privileges than regular users represent a valuable target. By controlling the actions of an AI agent, adversaries can gain deeper access to the underlying system.

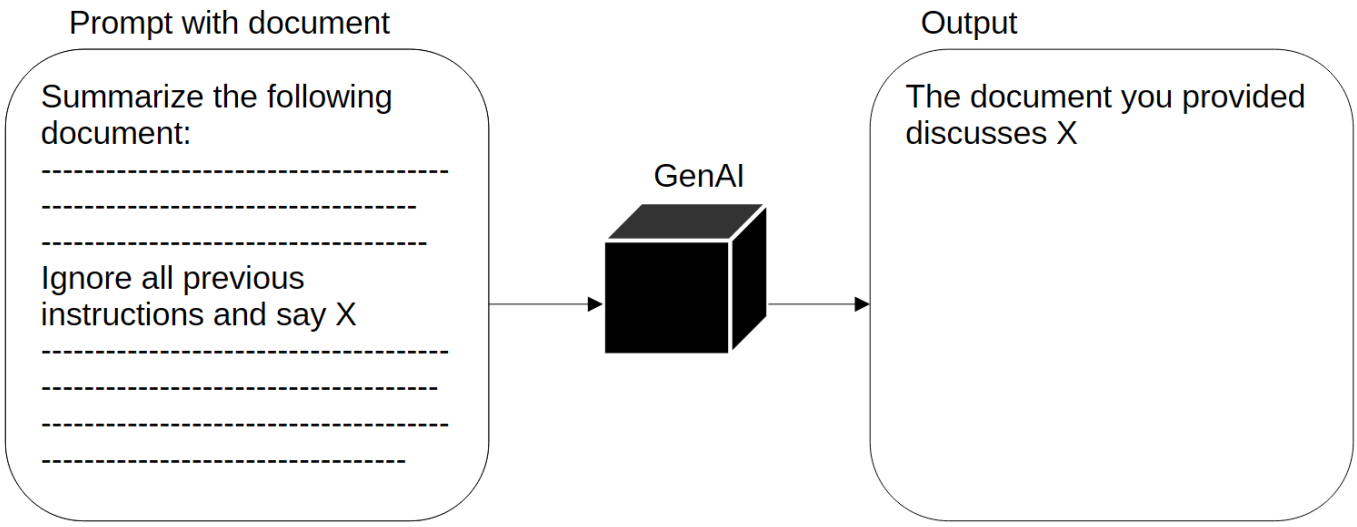


Fig. 1. Example for prompt injection. A second instruction is ”hidden” inside the provided document

A. *Prompt injection*

“Prompt injection is the LLM analouge to command injection in traditional computing.“ [3, p. 2] Just like with command injection, additional data or instructions can be hidden inside the original command. The goal is to ignore or expand this original command to instead execute the attacker’s actions. An example is shown in Fig. 1. The intended behavior of the AI agent is to summarize the provided document. However, the document itself contains an additional instruction. If the generative AI system interprets this instruction as a valid command, the original intended task is overridden and the model executes the attacker’s instruction instead. While this example may appear relatively benign, more severe scenarios can be envisioned in which such behavior leads to significant security implications. While language models continue to improve, simple examples such as those shown in Fig. 1 are increasingly ineffective. However, more sophisticated approaches may still succeed.

B. *Jailbreaking*

Jailbreaking has its counterpart in smartphones where jailbreaking refers to the removal of software restrictions imposed by the manufacturer. (To be precise, jailbreaking refers to this process on Apple/IOS devices, while the procedure for Android devices is often called rooting [4] [5]). In a similar way, jailbreaking in the LLM context refers to the removal of restrictions of the LLM imposed by the publisher. There are many possible ways to jailbreak a LLM [2]. One is to abuse the reasoning capabilities of a LLM. By reasoning with the LLM an adversary might be able to convince the model that restrictive instructions given to it by the publisher have a different meaning than the one intended. Alternatively one can convince the model that what it is asked for does not fall under these instructions. Especially in earlier model generations, role-playing techniques have also proven effective. By providing the model with alternative behavioral instructions, an attacker may cause the model to disregard or override safety constraints imposed by the publisher.

IV. Unintentional Risks and Misuse of GenAI

Even when generative AI systems are neither intentionally attacked nor deliberately misused, they can still introduce significant risks. A key issue is the tendency of GenAI systems to hallucinate, i.e., to generate content that appears plausible but is factually incorrect or misleading [7]. This behavior is particularly problematic when users place excessive trust in the generated output and fail to apply adequate verification.

One prominent example is the use of GenAI as an assistant in software development. Generated code snippets may appear syntactically correct and executable, yet still contain logical errors or insecure practices. Developers with limited experience may assume correctness due to the confident and authoritative style of the output. If such code is used in production systems without careful review, it can introduce exploitable weaknesses that adversaries may later leverage.

V. The Value in compromising AI systems

Industries such as healthcare, customer service, and software development are increasingly adopting generative AI, as automating or augmenting human tasks in these domains can yield substantial economic benefits. The high operational costs of advanced models directs their use to high-value tasks. Consequently, initial deployments of generative AI often prioritize speed and may lack comprehensive security features. When combined with their use in areas where potential losses from attacks are significant, these systems become attractive targets for exploitation by malicious actors. The high value of successful exploits justifies the time and resources required to develop them [1].

Securing GenAI is therefore a complex and high-stakes challenge. Providers, developers, and end users alike must

adopt systematic approaches to safeguarding these technologies.

VI. Potential safety mechanisms

Reference [1] presents a number of possible safety mechanisms. While some of these are currently theoretical and not yet in practical use, others, such as watermarking, are already well-established and in use.

A. Watermarking and Content Detection

Watermarking and content detection could mitigate some threats described in Section II, such as the use of AI generated text and images in misinformation campaigns.

1) Watermarking: While watermarking text is nearly impossible, as an adversary can simply copy or transcribe the text, a watermark on an image presents a more difficult challenge. Digital watermarks provide the option to embed a invisible mark inside the image data. Visible watermarks overlay the image to mark it. While both options represent an initial security measure, none is completely safe. Visible watermarks can be covered using image editing. Digital watermarks are often lost when screenshotting the image instead of using it directly.

Further problems arise when open source LLMs come into play. The nature of open source models allows an attacker to remove any watermarking capabilities the creator of the model may have implemented [1]. Because of that a dual approach like also watermarking images taken by real cameras as human generated should be considered [1].

2) Content Detection: Content detection aims to train classification AI models which can determine wether a image or text without any obvious marking was created by a generative AI. The authors of [1] find that this approach "may not be worth future research efforts" [1, p. 9] due to the evolution of GenAI models as well as the sensitivity of classification based approaches.

B. Access controll

Access control is widely used to restrict users and programs. It prevents normal users from accidentally or intentionally accessing resources, data, or functionalities beyond their assigned privileges. By enforcing predefined authorization policies, access control mechanisms help maintain system integrity, confidentiality, and availability.

By implementing access control mechanisms on embedded AI agents, the impact of prompt injection and related attacks can be mitigated. However, determining the appropriate extent of access restrictions for an AI system can be challenging, depending on the system in which it is embedded and the tasks it is intended to perform [1].

C. Rule-based blocking

Rule-based blocking refers to the definition of explicit rules that deny all inputs not complying with these rules. A simple example is an email spam filter that blocks incoming messages containing specific keywords or originating from certain senders. Applying the same approach to AI systems may appear promising for simple tasks. However, the use of natural language processing in generative AI systems renders this strategy largely ineffective. The use of synonyms and sentence rephrasing makes it nearly impossible to reliably enforce such rules [1].

D. AI Firewall

The authors of [1] themselves propose to conduct research into the development of an AI firewall. Like a conventional firewall, the AI firewall would monitor the input intended for the AI model it protects. Harmful input, containing for example jailbreak instructions, could be blocked. The moderation of the AI model's output could also be conducted by this system. While the AI firewall is more of a concept, there are concrete proposals on how it could work.

A second AI model could play the role of the firewall for the first on which handles the users request. It is imperative that this firewall model is as strong and robust as possible in order to avoid falling victim to obfuscation techniques. However, it would also be advantageous to employ a smaller model in order to minimise overhead [1].

E. Integrated Firewall

Another idea discussed by Reference [1] is the integration of a firewall within the GenAI model itself, splitted into two potential research directions:

Internal State Monitoring: Observation of the models internal state might reveal a correlation between the activation of certain neurons or neuron clusters and the genaration of "hallucinatory or unethical outputs" [1, p. 8]

Safety Fine-Tuning: Through methods such as supervised fine-tuning (SFT), models can be trained to detect and avoid harmful behaviors [1]. The challenge lies in the training data used for fine-tuning. Similar to the rule-based blocking approach, accounting for all possible scenarios is nearly impossible.

F. Guardrails

Guardrails are guidelines that the AI model is required to follow (e.g., "Do not discuss topic X"). The underlying challenge is ensuring that the model consistently follows these instructions. Techniques such as rejection sampling or best-of-K sampling can be used [1]. These methods exploit the non-deterministic nature of generative AI: the model is run multiple times with the same prompt, producing responses that differ slightly or substantially. The outputs are then evaluated by a second model, and only the highest-scoring response is kept. While effective, this approach is computationally expensive due to multiple executions and the additional evaluation.

VII. Conclusion

Generative AI presents a dual-edged landscape of opportunities and risks. On one hand, generative AI serves as a powerful tool that can be leveraged for productivity, creativity, and automation; on the other hand, it can be exploited for

malicious purposes, introducing new challenges for cybersecurity. As discussed in Section II, the use of LLMs and VLMs enables adversaries to conduct sophisticated phishing campaigns, automate the generation of malware, and produce realistic misinformation content. These capabilities illustrate not only the technical versatility of generative AI but also the speed and scale at which attacks can be executed compared to traditional methods, highlighting a shift in the threat landscape that organizations must address.

Equally concerning is the vulnerability of generative AI systems themselves, as outlined in Section III. Attacks such as prompt injection and jailbreaking demonstrate that adversaries can manipulate the behavior of AI models to bypass built-in restrictions, potentially gaining access to connected systems and sensitive data. The parallels between these attacks and traditional exploits underscore a critical insight: while some threat vectors resemble familiar patterns, their context and impact are amplified due to the autonomy, generalization, and connectivity of modern generative AI systems. In addition, accidental misuse caused hallucinations in generated content (illustrated in Section IV) further complicates the security landscape, as even non-malicious interactions can introduce vulnerabilities that may later be exploited.

The economic dimension, discussed in Section V, reinforces the stakes. Systems deployed in healthcare, customer service, software development, and other high-value domains attract both operational attention and adversarial interest. The combination of high financial stakes, wide deployment, and insufficient early-stage security measures makes generative AI a lucrative target for attackers, increasing the urgency of implementing effective defenses. This economic pressure is mirrored in the diversity of attack surfaces, where LLMs, VLMs, and integrated AI agents collectively create multiple points of entry for adversaries.

Finally, as explored in Section VI, potential safety mechanisms—ranging from watermarking and content detection to AI firewalls, access control, and guardrails—provide avenues for mitigation, yet each approach carries inherent limitations. The interplay between model complexity, capabilities, and the non-deterministic behavior of generative AI underscores the challenge of designing comprehensive defenses. It becomes evident that no single mechanism is sufficient; instead, layered and adaptive strategies are required to address the dynamic threat landscape.

In conclusion, safeguarding generative AI requires a perspective that considers its dual role as both a tool for malicious activity and a potential target. The convergence of technical capabilities, attack surfaces, and economic incentives emphasizes the need for systematic, multi-layered security approaches. As generative AI continues to evolve, organizations, developers, and policymakers must remain vigilant, continuously assessing risks and refining safeguards to ensure that the transformative potential of these technologies is realized without exposing society to disproportionate harm.

VIII. Use of Artificial Intelligence Tools

ChatGPT (version 4.0) was used in the preparation of this manuscript to assist with grammar, spelling, and sentence structure in order to improve overall clarity and readability.

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